# How Many Documents Cover a Corpus

**Anonymous ACL submission**

## Abstract

If we read 15 documents from a corpus, how much of the corpus have we seen? A survey sample comes with a sample size formula and a margin of error, and a sample of text comes with neither. We answer the question for one measure: the share of a corpus that has no document with the same content in the sample, where the same content means two embeddings closer than a distance $r$. A classical estimator due to Good, Turing and Maurer computes this share from the sample alone, as the share of sampled documents with no close match among the others. We show that on many corpora this estimator depends on the sample size $n$ and the distance $r$ only through $nr^d$, where $d$ is the intrinsic dimension of the corpus. This law yields a sample size formula, and in 77 of 83 tests the sample actually needed lay between two thirds and one and a half times the formula's answer. A pilot of 2,000 documents shows whether the law holds. With $r$ calibrated by large language models, 15 random posts show nearly every topic of a corpus that repeats a few claims, while samples of personal stories, forum questions and Truth Social posts cover little beyond themselves. On two corpora the answer depends on which model judged the pairs.

## 1 Introduction

Researchers who study large collections of text rarely read all of them. An annotation team codes a few thousand posts out of millions, a large language model summarizes a few hundred documents because the whole corpus does not fit in its context window, and a qualitative researcher keeps reading until further documents stop adding codes. Each of these practices assumes that the documents in hand stand for the documents left unread. Yet the number of documents is usually set by budget or by feel. If we read 15 posts from a corpus of 10,000, a number a person can read in a few minutes, have we understood the corpus? If not, would 100 posts be enough, or 1,000?

Survey research has a settled answer to the same question for a human population. The researcher states how much people differ on the quantity of interest (the population variance $\sigma^2$) and how large an error is tolerable ($e$), and the formula $n = z^2\sigma^2/e^2$ gives the number of people to interview (Cochran, 1977; Kish, 1965). Every estimate from the sample then carries a margin of error. A sample of a corpus comes with neither a sample size formula nor a margin of error. This paper builds both for the question a reader of a sample usually has in mind: how much of the corpus has a document like it among the documents we read?

The answer depends on what counts as alike. Two posts can be about the same topic, such as the 2020 United States Census, while making different claims, and two other posts can repeat the same claim in different words. We place each document in an embedding space, where a text encoder puts documents with similar meaning close together, and we measure how alike two documents are by the cosine distance between them. The user then chooses a distance $r$ below which two documents count as the same content. To give $r$ a concrete meaning, we asked two large language models, Qwen3.8 and Gemma 4, to judge about 4,000 pairs of documents at a range of distances. For each corpus we found the distance below which most pairs are about the same specific claim, question, event or story, and the larger distance below which most pairs share at least a topic (Section 5.6).

At these two distances the question of the 15 posts has a clear answer, and it differs from corpus to corpus (Table 3). Take the 17,202 tweets about the 2020 Census that contain the word "illegal". In a random sample of 15 of them, 98% of the corpus has a tweet on the same topic, so a reader of those 15 tweets has seen nearly every topic the corpus discusses. Only 38% of the corpus, however, has

a tweet in the sample that makes the same specific point. Now take 10,000 personal stories from the TIFU forum on Reddit, in which users tell how they messed something up. In a sample of 15 stories, only 1.5% of the corpus has a story on the same topic. Even in a sample of 1,000 stories, only 10% of the corpus has a story about the same event. Whether a sample is enough therefore depends on the corpus and on what the reader needs to know.

The share of a corpus that has no document like it in the sample is called the missing mass, and statistics has a classical way to estimate it without looking at the unread documents. During the Second World War, I. J. Good and Alan Turing noticed that a sample reveals how much it is missing through the items it contains only once (Good, 1953). If most words in a sample of text occur once, many more words remain unseen. If every word in the sample has occurred several times, few words remain. Maurer (2022) carried this idea from categories to distances. His estimator $G$, which we call the isolation rate, is the share of documents in the sample that have no other sampled document within $r$, that is, the share of the documents we read that were unlike anything else we read. For example, in a sample of 15 of the Census tweets with "illegal", 64% of the tweets have no close match among the other 14. This is the estimate. The truth, which we can compute because we have the whole corpus, is that 62% of the corpus has no close match in the sample. A reader can therefore learn how much a sample has missed from the sample alone. Maurer's paper is theoretical. It proves how far $G$ can be from the missing mass, and it leaves open how $G$ falls as the sample grows, which is what a researcher needs in order to plan a sample. To our knowledge the estimator has not yet been used on text.

How $G$ falls with the sample size follows from a classical result about the distances between random points (Levina and Bickel, 2005; Facco et al., 2017; Penrose, 2003). Suppose the documents of a corpus differ from one another along $d$ independent directions of meaning, where $d$ is called the intrinsic dimension of the embedded corpus and can be estimated from the distances between documents. Then the chance that a document has no neighbour within $r$ among $n$ sampled documents depends on $n$ and $r$ only through the product $nr^d$. In other words, a larger sample and a looser definition of the same content can stand in for each other, at a rate set by $d$. The product has a direct meaning for sampling. When the definition of the same content becomes twice as strict, that is, when $r$ is halved, the number of documents close enough to count as a match shrinks $2^d$ times, so the sample must grow $2^d$ times to cover the corpus as well as before. For the medical questions of the AskDocs forum and the TIFU stories, $d$ is about 13, so a definition twice as strict requires about 8,000 times as many documents. The intrinsic dimension thus does for a corpus what the variance does for a survey population. It is the property that sets how large a sample must be.

We test this law on seventeen pairs of corpus and encoder, which include medical questions, personal stories, election discussion, tweets about the 2020 Census, posts from Truth Social and Bluesky, and synthetic data with a known answer. The law holds on most of these corpora, it needs one correction for finite corpora, and it fails on a recognizable kind of corpus.

1. On ten of the seventeen pairs, $G$ depends on the sample size and the distance only through $nr^d$. The law has two constants besides $d$, and a fit of these constants on samples of up to 1,000 documents predicts $G$ for samples seven times larger. Solved for $n$, the law gives the sample size that brings $G$ down to a target, such as 20% of the sampled documents isolated. We tested this formula in 83 cases, one for each combination of corpus, target and distance at which the true answer lay beyond 1,000 documents. In 77 of the 83 cases, the sample that actually reached the target held between two thirds of the formula's number and one and a half times that number. A formula answer of 1,500 documents thus meant an actual need between 1,000 and 2,250. The standard tool from ecology for the same task, the extrapolation of Chao and Jost (2012), predicts $G$ for a sample twice as large with an average error two to nine times that of the law.

2. For a corpus of $N$ documents, the missing mass is $G\,(1-n/N)$. The correction is needed because $G$ describes the documents a sample has not reached, and the $n$ sampled documents are reached by definition. Survey sampling knows the same factor as the finite population correction. With it, the estimate is on average one percentage point from the true missing mass.

3. The law fails on the other seven pairs. Five are political corpora, among them a forum in which short slogans repeated many times sit beside longer discussion. One is a second encoding of that forum, and one is a synthetic mixture of topics. A test run on a pilot sample of 2,000 documents separates the seven failing pairs from the ten passing pairs before the full sample is drawn.

Taken together, these results let a researcher size a sample before reading it. We set a modest target, that half of the corpus should have a document with the same specific content in the sample. On corpora that repeat a few claims, such as Census tweets selected by a keyword or a forum about one election, a few hundred posts reach this target. On forum questions, personal stories and the Truth Social posts, the sample must hold more than 40% of the corpus. The reference point is a corpus in which no two documents share content. There the sample matches only itself, so reaching the target takes exactly half of the corpus, and the forum questions, stories and Truth Social posts come close to this limit. Two pilots of different sizes serve two purposes. A pilot of 100 documents already tells which kind of corpus a researcher has, since in it no document has a match on the corpora that come close to the limit. A pilot of 2,000 documents is needed to fit the law and predict the required sample.

## 2 Background

### 2.1 How much a sample has missed: the discrete case

The question of how much a sample has missed was first answered for samples of categories, such as words in a text or species in a forest. A sample of $n$ draws covers the categories it contains, and the missing mass is the total share of the population that belongs to categories the sample never saw. Let $f_k$ be the number of categories that appear exactly $k$ times in the sample. Good (1953) showed that $f_1/n$, the share of draws whose category appears only once, estimates the missing mass with little bias. The intuition is that a category seen once is the kind of category that could easily have been missed, so the number of such categories tells how much was missed. Gale and Sampson (1995) turned the estimator into a standard tool for estimating the frequency of rare words, and Good and Toulmin (1956) extended it to predict how much a larger sample would cover.

Ecologists built a whole sampling theory on the same idea, because a field survey of species faces our problem exactly: no survey finds every species, and the surveyor must decide when to stop. Chao (1984) bounded the number of species a survey has not found using $f_1$ and $f_2$, the counts of species seen once and twice. Chao and Jost (2012) then wrote the coverage a survey would reach after $m$ more draws as a formula in $n$, $f_1$ and $f_2$, so that a survey can be planned to stop at a target coverage. Ecologists now treat the curve of coverage against sample size as one object, extended backward to smaller samples and forward to larger ones (Colwell et al., 2012; Chao et al., 2014). The same ideas reach well beyond ecology. Efron and Thisted (1976) used them to estimate how many words Shakespeare knew but never used, and Orlitsky et al. (2016) showed how far ahead the number of unseen categories can be predicted, up to about $n \log n$ draws for a sample of $n$.

These estimators have been applied to text and to machine learning a few times. Hajič and Moss (2025) applied the Chao bound to a catalogue of chant to estimate how far the catalogue is from complete. Tran et al. (2017) fitted species models to the answers to an open ended survey question to predict when new kinds of answer stop appearing. Chung et al. (2018) counted the examples a training set is missing with the same estimator, treating each unique example as a species, and Pal et al. (2026) estimated how much of the situations a deployed classifier meets it has never seen in training. In all four studies an item belongs to a discrete category and two items either match or do not. Documents rarely fall into such categories. Two posts can make the same point in different words, and two posts with many words in common can make opposite points.

### 2.2 How much a sample has missed: the case of distances

Maurer (2022) removed the need for categories. He works in any space with a distance $\delta(\cdot, \cdot)$ between items. For a sample $X_1, \ldots, X_n$ and a distance $r$, the missing mass is the probability that a new item lies farther than $r$ from every item in the sample. Maurer calls it the conditional missing mass, because it depends on the sample actually drawn. His

estimator is

$$G(X, r) = \frac{1}{n} \left| \{k : \delta(X_k, X_i) > r \text{ for all } i \neq k\} \right|, \quad (1)$$

the share of sampled items that have no other sampled item within $r$. When items are categories, so that two items are at distance 0 if they share a category and at distance 1 otherwise, $G$ with $r < 1$ is exactly Good and Turing's $f_1/n$. Maurer proved that $G$ overstates the conditional missing mass by at most $1/n$ on average, and that the variance of the difference between the two is at most $3/n$ (his Theorem 2.1). His paper is theory. It sketches anomaly detection and data compression as uses, and it contains no data, no embeddings and no statement about how $G$ changes as the sample grows.

That change is described by a classical result about nearest neighbours. Suppose items are drawn from a density $f$ on a set of dimension $d$. Then the probability that an item at $x$ has no other item within $r$ among $n$ draws is close to $\exp(-nf(x)\, v_d\, r^d)$, where $v_d$ is the volume of a ball of radius 1 in $d$ dimensions (Evans et al., 2002; Penrose, 2003). The same result is the basis of the correlation dimension of Grassberger and Procaccia (1983) and of the estimators of intrinsic dimension proposed by Levina and Bickel (2005) and Facco et al. (2017), which work backward from nearest neighbour distances to $d$. We use it forward, to predict how $G$ falls as $n$ grows. The ecological work closest to ours is the attribute diversity of Chao et al. (2019). There, two species whose traits lie closer than a threshold $\tau$ count as partly the same species, and diversity is reported as a function of $\tau$. We treat $r$ the same way, as a resolution the user chooses for the question at hand.

### 2.3 Measures of diversity in embedding space

Several measures describe how spread out a set of embedded documents is, and none of them says how many documents a sample needs. The Vendi score (Friedman and Dieng, 2023) is an effective number of distinct items in a sample, computed from the eigenvalues of the matrix of cosine similarities among the items, and Pasarkar and Dieng (2024) extend it to a family of related measures. The precision and recall of Kynkäänniemi et al. (2019) count how many generated items fall inside the neighbourhoods of a set of reference items. Naeem et al. (2020) made these counts less sensitive to how spread out the reference set itself is, and MAUVE compares two sets of texts through the distributions of their embeddings (Pillutla et al., 2021). Each of these measures describes the sample in hand or compares two samples. None predicts what a larger sample would cover, and none has a parameter that plays the role of the population variance in a survey.

Two other lines of work use distances between embedded documents to choose documents. Core set selection picks the documents that make the largest distance from any document to the chosen set as small as possible (Sener and Savarese, 2018), and semantic deduplication removes documents that lie within a small distance of another document (Abbas et al., 2023; Lee et al., 2022). Both need the whole embedded corpus before they start, and neither says how many documents a target requires. The intrinsic dimension of learned representations has also been measured, for images (Pope et al., 2021), for the layers of deep networks (Ansuini et al., 2019), and for text, where it separates human writing from text generated by a model (Tulchinskii et al., 2023). We use the intrinsic dimension for a different purpose, as the property of a corpus that sets the sample size.

### 2.4 When to stop reading in practice

Where a sample size for text is set on purpose at all, it is set by a stopping rule. Qualitative researchers stop interviewing at saturation, when new interviews add no new codes. Guest et al. (2006) found that saturation came after about twelve interviews in their study, and a review of 23 empirical tests of saturation by Hennink and Kaiser (2022) places it between 9 and 17 interviews. De Paoli and Mathis (2024) carried the rule to coding done by a large language model, tracking how fast the count of unique codes grows as more text is coded. In topic discovery with large language models, TopicGPT reports that the number of topics it finds levels off after about 600 documents, and it stops when 200 further documents add no new topic (Pham et al., 2024). Maeda et al. (2025) fitted a topic model with eight topics to 522 open ended survey answers and found that a sample of 100 answers recovers more than 60% of the topics found in all 522, and a sample of 250 about 70%. Each of these studies observes one coverage curve and stops on it. None can say, before reading, where the curve will flatten for a new corpus. The present paper gives the curve a parameter, the intrinsic dimension $d$, and an estimator that a researcher can compute from the sample.

## 3 Method

### 3.1 What we estimate

A corpus is a finite set of $N$ documents $x_1, \ldots, x_N$. We embed each document with a fixed encoder, scale each embedding to length 1, and measure the distance between two documents by the cosine distance of their embeddings. Two documents with nearly the same meaning are at a distance near 0, and two unrelated documents are at a distance near 1. A sample $S$ is a set of $n$ documents drawn at random without replacement.

The user chooses a resolution $r$. Two documents within $r$ of each other count as the same content for the purpose at hand. A small $r$ asks for close matches, such as two posts that make the same claim, and a large $r$ accepts loose matches, such as two posts on the same topic. Embeddings from current encoders crowd into a narrow region of the space (Ethayarajh, 2019), so the same number, say $r = 0.2$, means different things for different encoders and corpora. Section 5.6 therefore ties $r$ to judgments of content.

The quantity we estimate is the missing mass of the sample at resolution $r$,

$$M(S, r) = \frac{1}{N} \left| \{ j : \delta(x_j, x_i) > r \text{ for all } i \in S \} \right|, \tag{2}$$

the share of corpus documents that have no sampled document within $r$. In words, $M$ is the share of the corpus that has nothing like it in the sample. If $M = 0.6$, then 60% of the corpus consists of documents for which the sample holds no document with the same content, and a summary or an annotation built from the sample can say nothing specific about them. Coverage is $1 - M$, the share of the corpus that does have something like it in the sample.

Computing $M$ requires the whole embedded corpus, because it compares every corpus document with the sample. The estimator $G$ of Equation 1 needs only the sample. For each sampled document it finds the nearest other sampled document, and it counts the sampled documents whose nearest sampled neighbour lies farther than $r$. We write $G(n, r)$ for its value on a sample of $n$ documents. Our target is the finite corpus in hand. A researcher samples because the corpus is too large to annotate or to place in a model's context, and in our experiments the whole corpus is available, so we can check every estimate against the true $M$.

### 3.2 How the isolation rate falls as the sample grows

The isolation rate $G$ is the share of sampled documents that are isolated from the rest of the sample. To plan a sample, we need to know how $G$ falls as $n$ grows, and how that depends on $r$.

Suppose the embedded documents lie close to a set of intrinsic dimension $d$. The intrinsic dimension counts the independent directions along which documents actually vary, which is far smaller than the 768 or 4,096 coordinates of an embedding. Around a document $x$, the share of the corpus that lies within $r$ is then about $c(x)\, r^d$, where $c(x)$ is larger where documents are densely packed. The chance that none of $n$ sampled documents falls within $r$ of $x$ is about $\exp(-c(x)\, n\, r^d)$. Averaging over documents gives

$$\mathbb{E}\, G(n, r) \approx \mathbb{E}_x \left[ \exp(-c(x)\, n\, r^d) \right] = \mathcal{L}_c(n\, r^d), \tag{3}$$

where $\mathcal{L}_c$ is the Laplace transform of the distribution of $c$ over documents. The argument of $\mathcal{L}_c$ is $nr^d$. Whatever the density looks like, $G$ depends on the sample size and the resolution only through this product, as long as one $d$ holds across the resolutions of interest. We call this the collapse law, because curves of $G$ against $n$ at different resolutions collapse onto one curve when they are plotted against $nr^d$.

The product tells a researcher how sample size and resolution trade against each other. Doubling the sample has the same effect on $G$ as loosening the resolution by $2^{1/d}$, which for $d = 13$ is only 5%. In the other direction, halving $r$ requires $2^d$ times as many documents, about 8,000 times for $d = 13$. Corpora with a high intrinsic dimension are therefore expensive to cover at a fine resolution, in the same way that populations with a large variance are expensive to survey precisely.

The shape of the curve depends on how much the density varies across the corpus. If every document sat in a region of the same density, $G$ would fall as a pure exponential in $nr^d$. Real corpora mix dense regions, where many documents say similar things, with sparse regions, where documents are unusual, and the curve then falls more slowly. We describe the shape by a stretched exponential, $G = \exp(-(\kappa\, n\, r^d)^b)$ with $0 < b \leq 1$, where a smaller $b$ means a more uneven density. Taking logarithms twice turns this into a straight line,

$$\log\big(-\log G(n, r)\big) = a + b \log\big(n\, r^d\big), \tag{4}$$

which we fit by least squares, with $d$ estimated beforehand.

We estimate $d$ with the TwoNN estimator of Facco et al. (2017), which uses, for every document, the ratio of the distance to its second nearest neighbour to the distance to its first. We average it over five pilot samples of 2,000 documents. To fit $a$ and $b$, we draw five independent samples at each of eleven sample sizes from 100 to 7,000 (no more than $0.7N$), and compute $G$ at twelve resolutions spaced evenly on a log scale between the 5th and the 99th percentile of the nearest neighbour distances in a pilot. Only cells with $0.02 < G < 0.98$ enter the fit, because the double logarithm is unstable near 0 and 1.

As a check on $d$ that does not use TwoNN, we also fit $\log(-\log G)$ on $\log n$ and $\log r$ as two separate regressors. If the collapse law holds, the coefficient on $\log r$ equals $b\,d$ and the coefficient on $\log n$ equals $b$, so their ratio is a second estimate of $d$. When the two estimates of $d$ disagree, no single dimension describes the corpus across the resolutions used.

### 3.3 The sample size formula

Solving Equation 4 for $n$ gives the sample size at which the isolation rate at resolution $r$ falls to a target $\varepsilon$,

$$n^*(\varepsilon, r) = \left( \frac{\log(1/\varepsilon)}{e^a} \right)^{1/b} r^{-d}. \qquad (5)$$

For example, $\varepsilon = 0.2$ asks for a sample in which at most 20% of the documents are isolated, which, by the argument of Good and Turing, is a sample that leaves about 20% of the corpus unseen. The corpus enters the formula through three numbers. The intrinsic dimension $d$ sets how fast the required sample grows as the resolution becomes finer. The constants $a$ and $b$ set the level and the shape. All three can be estimated from a pilot. Section 5.3 reports the formula with $a$ and $b$ fitted inside a pilot of 2,000 documents, as a user would fit them, and with $a$ and $b$ fitted on draws from the whole corpus.

### 3.4 The finite population correction

$G$ estimates the missing mass of the process that generated the corpus, as if new documents kept arriving. When the target is the corpus itself, two facts change the picture. The $n$ sampled documents are part of the corpus, and each of them has something like it in the sample, namely itself. Each of the other $N - n$ documents is isolated from the sample with a probability close to $G$. The missing mass of the corpus is therefore

$$\hat{M}(n, r) = G(n, r)\,\big(1 - n/N\big). \qquad (6)$$

If a researcher samples half of a corpus and $G$ is 0.8, then 80% of the unread half is unseen, which is 40% of the corpus. The factor $1 - n/N$ is the one survey sampling calls the finite population correction (Groves et al., 2009). In a survey it shrinks the variance of an estimated mean. Here it shrinks an expected share, for the simple reason that the documents in the sample are covered by definition.

### 3.5 A test for corpora with more than one scale

The collapse law assumes that one intrinsic dimension holds across the resolutions of interest, and a researcher needs to know whether that holds before drawing the sample. We therefore test the law inside a pilot of 2,000 documents. We fit Equation 4 on subsamples of the pilot with up to 750 documents, and we compute how much of the variance of $\log(-\log G)$ the fit explains on subsamples of 1,000 to 2,000 documents, which it did not see. A corpus passes when this pilot $R^2$ is at least 0.95. The two estimates of $d$ give a second test, since they should agree when the law holds.

To check whether the pilot test gives the right answer, we also fit the law on the whole corpus with samples of up to 1,000 documents and evaluate it on samples of up to 7,000. We call a corpus collapsing when this full out of sample $R^2$ is at least 0.9. When the law fails, two probes help find the reason. The first is the share of documents that have a near duplicate, a nearest neighbour within cosine distance 0.02. The second is the spread of the intrinsic dimension across twenty clusters of the corpus found by $k$-means, which shows whether different parts of the corpus vary along different numbers of directions.

### 3.6 Giving the resolution a meaning

A cosine distance means little to a reader until it is tied to content. For each real corpus, we drew about 40 pairs of documents in each of ten bins of distance between 0 and 1, about 400 pairs per corpus. Near pairs come from the nearest neighbours of random documents, and far pairs come from random pairs. A large language model read each pair and judged whether the two documents are about

the same specific content (the same claim, question, event or story), about the same general topic, or unrelated (Appendix J). The share of pairs judged the same content falls as the distance grows. We call $r_{\text{same}}$ the distance at which this share crosses one half, interpolated between the centres of neighbouring bins, and $r_{\text{topic}}$ the distance at which the share judged the same content or the same topic crosses one half. We obtain a 90% interval for each crossing by resampling the judged pairs within their bins 2,000 times. Two large language models judged every pair, Qwen3.8-27B and Gemma-4-31B-it, so that we can see how much the meaning of $r$ depends on the judge. We call them Qwen3.8 and Gemma in what follows. The encoder Qwen3-Embedding-8B, which we call Qwen3, is a different model from the same family.

## 4 Data

We test the law on two kinds of data. Synthetic data have a known answer, so they show whether the method recovers the truth. Real corpora show whether the law holds on text that researchers actually sample. Table 1 lists all of them.

The synthetic data come in two families. The Gaussian manifolds place 20,000 points on a random subspace of dimension $d \in \{2, 4, 8\}$ inside a space of 64 dimensions, so the true intrinsic dimension is known. The Zipf mixture draws 20,000 points around 200 topic centres, with topic sizes that follow a Zipf distribution (a few large topics and many small ones), adds Gaussian noise, and scales each point to length 1, so the topic of every point is known.

The ten real corpora come from four sources and cover several kinds of text. Three forum corpora from Reddit were assembled for earlier work on summarizing corpora. AskDocs holds medical questions, Trump holds posts about the 2020 presidential election, and TIFU holds personal stories told in the first person. Each has 10,000 documents after cleaning and the removal of duplicates. We embed them twice, with GTR-T5-base (Ni et al., 2022) on three random samples of the source (seeds 42 to 44), and with Qwen3-Embedding-8B (Zhang et al., 2025) on the first of these samples. In the tables, the three GTR samples of a forum corpus count as one pair of corpus and encoder, so the three forum corpora give six pairs. With the five Census slices, Truth Social, Bluesky and the four synthetic sets, this makes the seventeen pairs of Table 1.

Five corpora come from a 2020 collection of tweets about the United States Census. Each is a slice selected by a keyword (citizens, trump, covid, illegal) or, for the random slice, drawn without a keyword. After removing links and mentions and dropping tweets with identical text, the slices hold between 7,610 and 30,136 tweets.

Two corpora are random samples of 50,000 English posts each from Truth Social and Bluesky, written between May and December 2024. We collected the public posts through the platforms' public interfaces. For the samples we kept posts marked as English with at least 20 characters after links and mentions were removed, and we dropped posts with identical text, which also removes reposts that copy their original. The Truth Social collection was built from a keyword search for "trump". It holds the matching posts, the threads under them, and the timelines of many of their authors. The Bluesky collection was built from a search for election terms, such as the names of candidates and parties, "election" and "politics", and it also holds many posts that match none of these terms, added by a crawl whose rule for adding posts was not recorded. Neither collection is a sample of its platform.

All Census, Truth Social and Bluesky corpora are embedded with Qwen3-Embedding-8B at 4,096 dimensions, with text cut at 256 tokens and each embedding scaled to length 1. We use cosine distance throughout. Cosine distance is not a metric in the strict sense, but it orders neighbours exactly as the metric $\sqrt{2(1-\cos)}$ does, so every nearest neighbour in this paper is the same under either.

The posts are public, and the analysis uses only their embeddings and statistics computed over whole corpora. No post or account is quoted or identified in the paper. We release the post identifiers of the Truth Social and Bluesky samples, the pair judgments and the code, and no post text.

## 5 Results

### 5.1 The isolation rate tracks the true coverage

Before testing the law, we check that $G$ measures what it should. In the synthetic Zipf mixtures every document belongs to a known topic, so we know exactly which share of the topics a sample has covered. Over four settings of the mixture, three random seeds and eleven sample sizes from 50 to 5,000, the coverage $1 - G$ at a resolution of

half the median distance between documents in a pilot follows the true topic coverage closely. The rank correlation is $\rho = 0.968$, and $1 - G$ is off by 2.9 percentage points on average (Appendix A). Good and Turing's estimator applied to the true topic labels, which $G$ never sees, does only slightly better ($\rho = 0.989$). At smaller resolutions the agreement falls away. At those resolutions two documents of the same topic often lie farther apart than $r$, so $G$ counts a document as unseen content even when its topic is in the sample. The resolution must therefore match the question being asked of the sample, and Section 5.6 ties it to judgments of content.

### 5.2 The isolation rate collapses on $nr^d$

Table 1 reports the fit of Equation 4 on 17 pairs of corpus and encoder, and Figure 1 shows six of them. On ten pairs the law holds: the three synthetic manifolds, AskDocs and TIFU under both encoders, the random slice of the Census tweets, Truth Social and Bluesky. On these ten, a fit on samples of up to 1,000 documents explains 92% or more of the variance of $\log(-\log G)$ in samples of up to 7,000 documents. A fit on samples of up to 300 documents still explains more than 92%, although it predicts samples 23 times larger than any it saw. In terms of $G$ itself, the law predicts the isolation rate of the largest sample with an average error between 1.5 and 4.2 percentage points on these ten pairs (Table 1). A researcher who has drawn a few hundred documents can thus predict from them alone what share of a sample twenty times larger will be isolated, at every resolution at once.

In each panel of Figure 1, the points of one colour are the values of $G$ at one resolution, for sample sizes from 100 to 7,000. For AskDocs and TIFU, the points of all colours lie on one curve once the horizontal axis is $nr^d$, so that a larger sample and a looser resolution are interchangeable. For Trump and for the Zipf mixture, each colour traces its own curve, and no single $d$ brings them together.

The second estimate of $d$, from the separate coefficients on $\log n$ and $\log r$, agrees with the TwoNN estimate where the law holds. On the ten passing pairs the two estimates differ by at most 19%. On the seven failing pairs they differ by 27% to 43%, and by 46% on one of the three GTR samples of Trump, and the regression estimate is always the lower one.

The law also predicts better than the two obvious alternatives. Both alternatives use the same samples of up to 1,000 documents to predict $G$ at the largest sample. The first alternative fits a separate curve in $n$ at each resolution, so it needs no $d$. On nine of the ten passing pairs its average error is 1.5 to 3.8 times that of the collapse law, and on the random Census slice it does better (an average error of 2.5 against 4.0 percentage points). The law wins because it pools the information from all resolutions through one $d$, while the separate curves each use one resolution. The second alternative is the extrapolation of Chao and Jost (2012), the standard ecological tool, which predicts coverage from the counts of singletons and doubletons. We take a singleton to be a sampled document with no other sampled document within $r$, and a doubleton to be one with exactly one. Run within its recommended range, from 1,000 to 2,000 documents, its average error on the passing pairs is 2 to 9 times that of the collapse law. Both alternatives beat the law on the Zipf mixture, which fails the pilot test.

### 5.3 The sample size formula

To test the formula, we treated each pair of corpus and encoder as a researcher would. We fitted Equation 4 on samples of up to 1,000 documents, and Equation 5 gave the sample size $n^*$ needed to bring the isolation rate down to 30%, 20% or 10% at each resolution. A case is one combination of corpus, target and resolution. We kept every case in which the true sample size lies beyond 1,000 documents, so that each case is a genuine prediction, and compared $n^*$ with the sample size at which $G$ actually reached the target in repeated draws. The ten passing pairs give 83 such cases, between three and seventeen per pair (Appendix B).

On the ten passing pairs, the formula comes close in 77 of 83 cases, in the following sense: the sample that actually reached the target held between two thirds of $n^*$ and one and a half times $n^*$. A formula answer of 1,500 documents thus meant an actual need between 1,000 and 2,250 documents. When $a$ and $b$ are fitted inside one pilot of 2,000 documents, as a user would fit them, with $d$ averaged over five pilots, the formula comes close in 76 of 83 cases. On the seven failing pairs it comes close in only 10 of 23 cases, and in 10 of 18 if the Zipf mixture is left out (Appendix B).

To see how much a single value of $n^*$ can be trusted, we drew 50 pilots, refitted $d$, $a$ and $b$ on each, and took the spread of $n^*$ across pilots as

an interval. These intervals are narrow, with a median width of 19% of $n^*$, and they contain the true sample size in only 21% of the cases. (This check used its own random draws, in which two of the 83 true sample sizes fell below 1,000, so it covers 81 cases.) The error of $n^*$ therefore comes mostly from the shape of the law, which is an approximation, and only a little from which pilot was drawn. We report $n^*$ with the band observed in the tests instead: the true need lies between two thirds and one and a half times $n^*$. The band contains the true sample size in 93% of the cases on passing corpora. On every passing corpus it contains most of the cases, and the worst corpus is Bluesky, with three of its five cases inside the band.

### 5.4 The pilot test

The pilot test must work before the sample is drawn, so we compute it inside one pilot of 2,000 documents. The ten passing pairs reach a pilot $R^2$ of 0.96 or more, and the seven failing pairs reach at most 0.93 (Table 1). A threshold of 0.95 on the pilot therefore sorts every pair the same way as the full test on the whole corpus. We chose the threshold on these corpora, so it should be confirmed on others. The agreement of the two estimates of $d$ gives the same sorting independently: the two differ by at most 19% on every passing pair and by more than a quarter on every failing pair.

The law should fail when the effective dimension of a corpus changes with the resolution, and the probes show such a change on Trump (Table 7 in Appendix E). Its twenty $k$-means clusters differ widely in local dimension. Clusters with a dimension near 2 hold slogans, short posts repeated in many small variations, and clusters with a dimension near 16 hold discussion. At small resolutions the isolated documents come mainly from one kind of cluster, and at large resolutions mainly from the other, so no single $d$ lines up the curves. Removing the 3.4% of documents that have a near duplicate leaves the failure in place. The Zipf mixture fails for a related reason: at small resolutions its documents look like noise around 200 topic centres, and at large resolutions they look like 200 points. The probes do not separate the failing Census slices from the passing corpora. Our best guess is that these slices were selected by a keyword that often appears in slogans and calls to action, but we have no direct evidence for it.

### 5.5 The finite population correction

The correction matters as soon as a sample is a noticeable share of the corpus. We compared $G$ and $G\,(1-n/N)$ with the true missing mass $M$ on the eight forum and Census corpora under Qwen3, at sampling fractions from 1% to 70% of the corpus and three resolutions, 144 cases in all (Table 8 in Appendix F). Without the correction, $G$ is off by 11 percentage points on average and by as much as 49 points. With it, the estimate is off by 1.2 points on average and by at most 11 points. When half of a corpus is sampled, the uncorrected $G$ reports about twice the unseen share that is actually there (0.40 against 0.20 in the average over corpora). On the three GTR forum corpora and the Zipf mixture the correction lowers the average error from 6.8 to 1.5 points. The correction holds on the corpora that fail the collapse law, because it only uses the fact that sampled documents are covered.

### 5.6 What a distance means

On every corpus, the share of pairs that the judge calls the same content falls steadily as the distance grows. Where it crosses one half, however, depends on the corpus (Table 2). Under Qwen3.8, $r_{\text{same}}$ lies at 0.05 or below on TIFU and reaches 0.28 on the Census tweets that contain the word "illegal". On Bluesky fewer than half of even the closest pairs are judged the same content, so no distance qualifies. Long personal stories differ in their details even when their embeddings are close, while short posts that repeat a slogan remain the same content at a larger distance. The distance at which two documents share a topic varies just as much, from 0.33 on TIFU to 0.60 on AskDocs. On the Census tweets that contain the word "citizens", every judged pair closer than 0.4 shares at least a topic, so the whole slice is in effect one topic. A cosine distance therefore has no fixed meaning across corpora, and a user who wants $r$ to mean "the same claim" has to judge a few hundred pairs from the corpus at hand.

A second model, Gemma-4-31B-it, judged the same pairs with the same prompt. The two models give the same label to 71% of pairs (Cohen's $\kappa = 0.55$, and $\kappa = 0.68$ for the same content against the other two labels). Gemma calls more pairs the same content than Qwen3.8 does, 39% against 28%, so its crossing lies farther out on every corpus. Nine corpora have a crossing under both judges, since Bluesky has none. On six of these nine the 90%

intervals of the two crossings do not overlap. The choice of judge therefore moves $r_{\text{same}}$ more than the sampling of pairs does. The judge is therefore part of the definition of $r$, and a user should report it together with the encoder.

### 5.7 What a reader of a few documents has seen

With the resolution tied to content, we can return to the question of the introduction. Table 3 gives, for random samples of 15, 100 and 1,000 documents, the share of the corpus that has no document with the same content in the sample, and the share that has no document even on the same topic. Each value is the true missing mass, computed against the whole corpus and averaged over 20 samples, with the distances set by the judgments of Qwen3.8.

The table shows two groups of corpora. On TIFU, AskDocs, Truth Social and the random Census slice, a sample covers almost nothing at the level of specific content. Fifteen documents leave more than 99% of the corpus with nothing like it in the sample, and 1,000 documents still leave between 84% and 96%. The documents of these corpora rarely share specific content, so each sampled document stands only for itself. At the level of topic the same corpora differ sharply. Fifteen questions from AskDocs leave 58% of the corpus without a question on the same topic, and 100 questions leave 15%. Fifteen stories from TIFU leave 98.5% without a story on the same topic, and even 1,000 stories leave more than half, because in the judge's view two personal stories rarely share a topic.

The second group, the Trump forum and the Census slices selected by a keyword, behaves differently. Fifteen posts already show the reader almost every topic, leaving between 1.8% and 11% of the corpus without a post on the same topic. At the level of specific content, 15 posts leave between 62% and 90% of the corpus unseen. A sample of 1,000 posts still leaves between 16% and 42%. These corpora repeat a limited number of claims, so a sample of moderate size contains a large part of what is said.

The answer to the question of the introduction is therefore conditional. Fifteen documents give a fair picture of the topics of a corpus that repeats a few claims, a poor picture of the topics of a corpus of personal stories, and a poor picture of the specific content of any corpus we studied. A reader does not need the whole corpus to learn which case applies. Once $r$ is set, the estimate $G\,(1 - n/N)$ computed from the 15 sampled documents alone, averaged over the same 20 samples, lies within 6 percentage points of the true share on every corpus and at both levels, and within 1.6 points on average. With 100 documents it lies within 2 points. The test is easy on the corpora where nearly everything is unseen, since there $G$ is almost always 1. It is informative on the repetitive corpora, where $G$ falls well below 1 and still follows the truth.

### 5.8 What a coverage target costs

A researcher who wants a sample to stand for most of a corpus at the level of specific content can ask how large the sample must be. We take a modest target, that half of the corpus should have a document with the same content in the sample. This target concerns the corpus, through the corrected missing mass $G\,(1 - n/N)$, while the formula of Section 5.3 targets the isolation rate $G$ of the sample itself. Table 2 gives the random sample that reaches this target on each corpus and under each judge. We measured it on the whole corpus, as the sample size at which $G(n, r_{\text{same}})\,(1 - n/N)$ falls to one half, with three random samples at each size up to $0.7N$. A useful reference point is a corpus in which no two documents share content. In such a corpus the sample covers only itself, so reaching the target takes exactly half of the corpus.

The corpora fall into the same two groups as in Table 3. On the Trump forum and the Census slices with the keywords citizens, trump and illegal, which repeat a few claims, at most 604 documents reach the target under either judge. On AskDocs, TIFU and Truth Social the sample must hold between 41% and 50% of the corpus, close to the 50% of a corpus with no shared content at all. For these corpora, a sample covers what it contains and little more, and an analysis that needs every specific claim in the corpus has to read the corpus. At the calibrated distance, a pilot of only 100 documents tells the two groups apart. In a pilot of 100, the isolation rate $G$ is 1.00 on AskDocs, TIFU and Truth Social, so no sampled document has a match in the pilot. On the repetitive corpora it is at most 0.90.

Two corpora depend on the judge. The Census slice with the keyword covid needs 19% of the corpus under Qwen3.8 and 1.6% under Gemma. The random Census slice, drawn without any keyword, needs 44% under Qwen3.8 and 7% under Gemma. Under the stricter judge these corpora look like collections of distinct documents, and under the looser

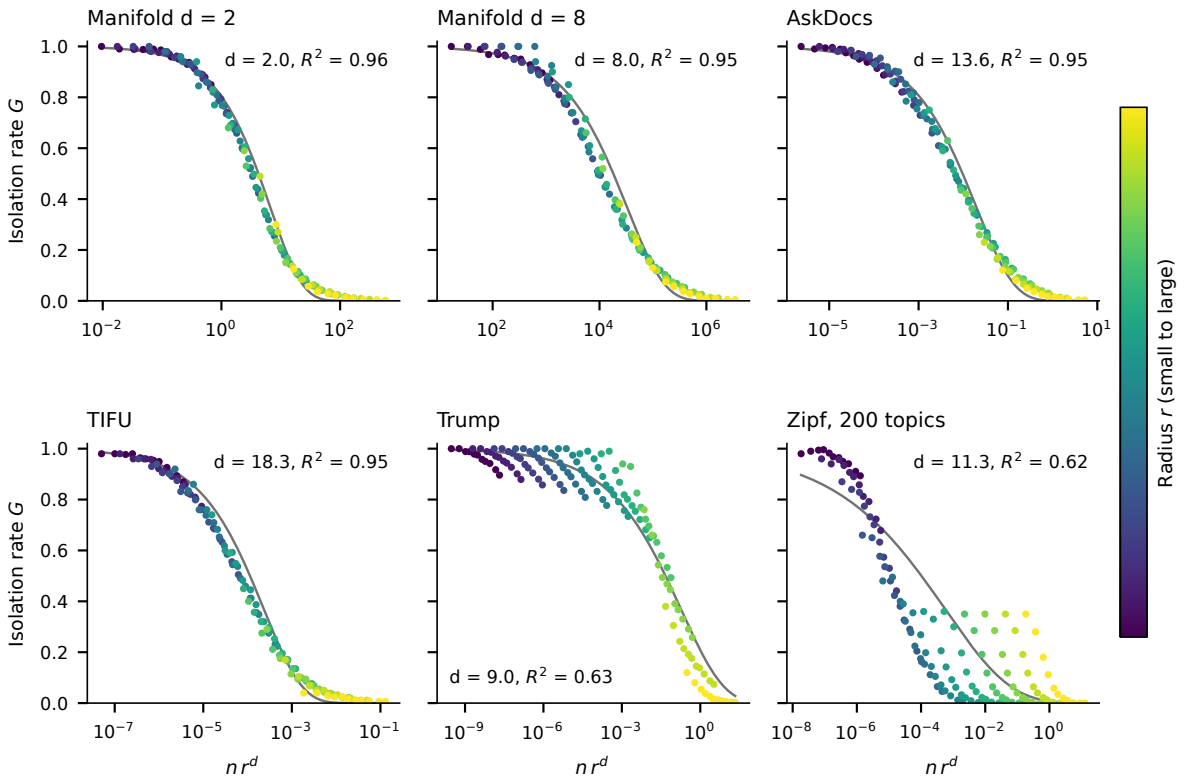



Figure 1: Isolation rate $G$ against $nr^d$ (log scale), one panel per corpus, with the GTR encoder for the three forum corpora. Each point is one sample size at one resolution, and the colour gives the resolution, from the 5th (dark) to the 99th (light) percentile of the nearest neighbour distances in a pilot. Where the law holds (the synthetic manifolds, AskDocs, TIFU), points from all resolutions fall on one curve. Where it fails (Trump, the Zipf mixture), the points of each resolution form their own curve. The grey line is the fit of Equation 4 on samples of up to 1,000 documents. Each panel gives $d$ and the full out of sample $R^2$ from Table 1.

judge they look repetitive.

The collapse law fitted on a pilot can predict this sample size before the corpus is embedded. We fitted $d$, $a$ and $b$ on one pilot of 2,000 documents, computed $G(n, r_{\text{same}})$ for every $n$ from Equation 4, applied the correction $1 - n/N$, and found where the result falls to one half. The prediction puts every corpus in the same group as the measurement. How close it comes depends on where the target is reached (Appendix C). When the target is reached below 100 documents, the prediction is below 100 too. When it is reached at the 50% limit, as on TIFU and AskDocs, the prediction is right almost by necessity, and these cases test little. When it is reached between 100 and 1,000 documents, the predicted sample is at most a third larger or a quarter smaller than the measured one. On the Census trump slice, for example, the law predicts 472 documents and the measurement gives 604. In the four cases that lie beyond the pilot's range, the prediction is 8% too high on the Census covid slice. On Truth Social and the random Census slice the measured share is 41% to 44%. There the prediction is about 4 percentage points too high. The law thus places these corpora correctly near the limit, and it places them a little too close, so it advises a slightly larger sample than needed.

The distance $r_{\text{same}}$ can also be found inside the pilot, so that the whole procedure runs on 2,000 documents. On five corpora, Gemma judged up to 400 pairs drawn within the first pilot, and $d$, $a$ and $b$ came from the same pilot (Appendix D). On the random Census slice, Truth Social, the Trump forum and AskDocs, the 90% interval of the pilot's crossing contains the crossing found with pairs from the whole corpus, and the share predicted from the pilot alone is within 2.5 percentage points of the measured share. In TIFU, the pilot of 2,000 stories contains no pair closer than 0.15, so it cannot locate a crossing that lies below that distance. It can only say that the crossing, if there is one, lies below 0.15, and at that bound the law predicts that half of the corpus is needed, as measured. TIFU and AskDocs sit at the 50% limit, where the answer

| Corpus | Encoder | $N$ | $d$ (sd) | $d$, ratio | $b$ | $R^2$ pilot | $R^2$ full | error, law | error, per resolution |
|---|---|---|---|---|---|---|---|---|---|
| AskDocs | Qwen3 | 10,000 | 12.8 (0.39) | 12.2 | 0.54 | 0.98 | 0.99 | 0.015 | 0.057 |
| Truth Social | Qwen3 | 50,000 | 12.6 (0.15) | 14.4 | 0.43 | 0.98 | 0.98 | 0.025 | 0.053 |
| Census random | Qwen3 | 7,610 | 12.9 (0.39) | 12.4 | 0.27 | 0.98 | 0.97 | 0.040 | 0.025 |
| Manifold $d$=2 | none | 20,000 | 2.0 (0.05) | 2.4 | 0.80 | 0.97 | 0.96 | 0.027 | 0.065 |
| TIFU | Qwen3 | 10,000 | 13.6 (0.35) | 13.4 | 0.53 | 0.98 | 0.96 | 0.039 | 0.086 |
| Bluesky | Qwen3 | 50,000 | 16.0 (0.69) | 15.2 | 0.42 | 0.98 | 0.96 | 0.037 | 0.056 |
| TIFU | GTR | 10,000 | 18.3 (0.33) | 19.0 | 0.50 | 0.97 | 0.95 | 0.040 | 0.070 |
| Manifold $d$=8 | none | 20,000 | 8.0 (0.21) | 8.1 | 0.62 | 0.97 | 0.95 | 0.042 | 0.081 |
| AskDocs | GTR | 10,000 | 13.6 (0.36) | 13.6 | 0.51 | 0.98 | 0.95 | 0.042 | 0.087 |
| Manifold $d$=4 | none | 20,000 | 4.0 (0.16) | 4.3 | 0.72 | 0.96 | 0.92 | 0.036 | 0.091 |
| Census illegal | Qwen3 | 17,202 | 11.4 (0.33) | 7.5 | 0.26 | 0.93 | 0.85 | 0.026 | 0.053 |
| Census covid | Qwen3 | 17,837 | 10.7 (0.40) | 7.2 | 0.30 | 0.85 | 0.79 | 0.045 | 0.073 |
| Trump | Qwen3 | 10,000 | 9.0 (0.51) | 5.3 | 0.26 | 0.90 | 0.78 | 0.045 | 0.075 |
| Census trump | Qwen3 | 30,136 | 9.0 (0.84) | 6.3 | 0.26 | 0.87 | 0.78 | 0.043 | 0.056 |
| Census citizens | Qwen3 | 7,849 | 9.1 (0.24) | 6.6 | 0.27 | 0.76 | 0.73 | 0.043 | 0.058 |
| Trump | GTR | 10,000 | 9.0 (0.48) | 5.2 | 0.27 | 0.79 | 0.63 | 0.054 | 0.069 |
| Zipf, 200 topics | none | 20,000 | 11.3 (0.35) | 6.4 | 0.21 | 0.66 | 0.62 | 0.136 | 0.026 |

Table 1: The collapse law on 17 pairs of corpus and encoder, sorted by the full out of sample $R^2$. Pairs above the line pass the pilot test. $d$: intrinsic dimension by TwoNN, mean and standard deviation over five pilots of 2,000 documents (over three samples of the source for the GTR forum corpora). $d$, ratio: the second estimate of $d$, the ratio of the coefficients on $\log r$ and $\log n$. $b$: shape exponent of Equation 4, fitted on samples of up to 1,000 documents. $R^2$ pilot: share of variance explained on subsamples of 1,000 to 2,000 documents of one pilot, by a fit on subsamples of up to 750. $R^2$ full: share of variance explained on samples of up to 7,000 documents, by a fit on samples of up to 1,000. Error, law and error, per resolution: average error in predicting $G$ at the largest sample, for the collapse law and for a separate curve in $n$ at each resolution.

hardly depends on $r$, so they test the calibration inside the pilot only weakly. The random Census slice, Truth Social and the Trump forum carry the test.

## 6 Discussion

The question we started from, whether 15 documents are enough to understand a corpus, has no single answer. How much a sample covers depends on two things the reader rarely states. The first is what counts as the same content, which we made explicit as the resolution $r$. The second is a property of the corpus, its intrinsic dimension $d$, which sets how fast coverage grows with the sample. A corpus of tweets that repeat a few claims and a corpus of personal stories can both hold 10,000 documents, yet a reader of 15 of the tweets has seen most of their topics, while a reader of 1,000 of the stories has still seen little of the stories. A sample size that worked in one setting, such as the dozen interviews after which qualitative researchers often stop, or the few hundred documents a large language model can take in one prompt, therefore carries no guarantee for another corpus.

For a researcher who samples text, the results suggest a short procedure. First, decide what the sample must cover, and give the resolution a meaning by having a few hundred pairs of pilot documents judged, by people or by a large language model, at a range of distances (Section 5.6). Second, embed a pilot of about 2,000 documents, estimate $d$ with TwoNN, and fit Equation 4 on subsamples of the pilot. Third, check the fit with the pilot test. If the pilot $R^2$ is at least 0.95 and the two estimates of $d$ agree, take the sample size from Equation 5, and expect the true need to lie between two thirds and one and a half times that number. If the test fails, the corpus has more than one scale, and the fit should be trusted only for sample sizes inside the range the pilot covers. Fourth, after the sample is drawn, report $G\,(1 - n/N)$ as the share of the corpus that has nothing like it in the sample. Once $r$ is set, this last step needs no pilot fit, and it already works on 15 documents.

The comparison with survey sampling can be made at three points. The factor $1 - n/N$ is the same finite population correction, applied to an expected share instead of a variance. The intrinsic dimension $d$ plays part of the role of the population variance. Like the variance, it is a property of the population, it is estimated from a pilot, and it sets how fast the required sample grows as the target becomes stricter. The margin of error is the third place, and here the two settings differ. A survey's margin of error bounds the error of an estimated

| | | Qwen3.8-27B | | | | Gemma-4-31B | | | |
|---|---|---|---|---|---|---|---|---|---|
| Corpus | Diag. | $r_{\text{same}}$ [90%] | $n$ (share) | pred. | interval | $r_{\text{same}}$ [90%] | $n$ (share) | pred. | interval |
| TIFU | pass | $\leq$.05 [.05, .10] | 4,997 (50%) | 50% | 50% | .08 [.05, .10] | 4,997 (50%) | 50% | 50% |
| AskDocs | pass | .15 [.13, .16] | 4,953 (50%) | 50% | 50% | .17 [.16, .19] | 4,923 (49%) | 50% | 49% |
| Truth Social | pass | .16 [.15, .18] | 21,942 (44%) | 48% | 42% to 45% | .18 [.17, .20] | 20,610 (41%) | 45% | 39% to 43% |
| Cens. random | pass | .12 [.10, .18] | 3,352 (44%) | 48% | 35% to 45% | .28 [.26, .29] | 532 (7.0%) | 9.2% | 3.9% to 11% |
| Cens. covid | fail | .20 [.19, .21] | 3,356 (19%) | 20% | 17% to 22% | .29 [.22, .30] | 279 (1.6%) | 1.7% | 0.6% to 14% |
| Cens. citizens | fail | .23 [.20, .26] | 370 (4.7%) | 4.9% | <1.3% to 10% | .35 [.32, .38] | <100 (<1.3%) | 0.1% | <1.3% |
| Trump (forum) | fail | .24 [.22, .26] | 442 (4.4%) | 4.1% | 2.2% to 10% | .31 [.31, .33] | <100 (<1.0%) | 0.4% | <1.0% |
| Cens. trump | fail | .24 [.17, .25] | 604 (2.0%) | 1.6% | 1.0% to 18% | .31 [.29, .33] | <100 (<0.3%) | 0.2% | <0.3% |
| Cens. illegal | fail | .28 [.25, .29] | <100 (<0.6%) | 0.3% | <0.6% | .30 [.29, .32] | <100 (<0.6%) | 0.1% | <0.6% |

Table 2: The distance that means the same content, and the random sample after which half of the corpus has a document with the same content in the sample. $r_{\text{same}}$: distance at which half of the judged pairs are about the same specific content, with a 90% interval from resampling the pairs. $n$ (share): the sample size, measured on the whole corpus, as a count and as a share of the corpus. The share can never exceed 50%, because the sampled half of a corpus is covered by itself. pred.: the share predicted from a pilot of 2,000 documents alone. interval: the measured share at the two ends of the 90% interval of $r_{\text{same}}$. <100: reached before the smallest sample of 100. $\leq$: the crossing lies at the first bin, and 32% of resamples find no crossing. Bluesky passes the pilot test but has no crossing, although 26% (Qwen3.8) and 34% (Gemma) of resamples find one. Truth Social and Bluesky are samples of 50,000 posts, and their shares describe corpora of that size. Diag.: the pilot test of Section 5.4.

| | Same content | | | Same topic | | |
|---|---|---|---|---|---|---|
| Corpus | 15 | 100 | 1,000 | 15 | 100 | 1,000 |
| TIFU | 99.8 | 99.0 | 90.0 | 98.5 | 91.5 | 57.5 |
| AskDocs | 99.8 | 99.0 | 89.9 | 58.3 | 15.4 | 0.7 |
| Truth Social | 99.9 | 99.6 | 96.4 | 63.9 | 32.1 | 10.0 |
| Census random | 99.7 | 98.2 | 84.0 | 15.8 | 7.0 | 2.1 |
| Census covid | 98.0 | 91.6 | 70.2 | 2.3 | 0.4 | 0.0 |
| Census citizens | 89.9 | 67.8 | 33.1 | | | |
| Trump (forum) | 88.0 | 68.0 | 40.2 | 3.6 | 1.0 | 0.1 |
| Census trump | 86.9 | 66.7 | 42.1 | 10.8 | 4.3 | 0.9 |
| Census illegal | 62.5 | 35.1 | 15.9 | 1.8 | 0.6 | 0.2 |

Table 3: The percentage of the corpus that has nothing like it in a random sample of 15, 100 or 1,000 documents, at the distance where half of the judged pairs are the same content ($r_{\text{same}}$) or at least the same topic ($r_{\text{topic}}$), both judged by Qwen3.8. True values computed against the whole corpus, averaged over 20 samples. The Census citizens slice has no topic distance, because nearly every pair in it shares the topic.

proportion. For proportions, the plain share in a random sample already estimates the corpus share, within 0.3 to 7 percentage points for the shares of the fifty clusters we tested (Appendix I). A random sample does not, however, promise to contain an example of each kind of content. The missing mass measures this risk, which matters most when the task is to describe or list what a corpus contains, as a summary or a qualitative codebook does. A report of a corpus sample can carry both numbers, the margin of error for proportions and the missing mass for content.

Coverage can also be measured against a list of themes, and this gives a different quantity. When themes divide the corpus into parts, the expected share of the theme mass that a sample contains depends only on how large the themes are, through $\sum_k m_k(1-(1-m_k)^n)$ with $m_k$ the share of theme $k$. This expression matched 1,200 samples from eight corpora to within 0.4 percentage points on average (Appendix G). Theme coverage therefore depends on the sizes of the themes in the list the researcher writes down. A list of a few large themes is covered quickly, and a list of many small themes slowly. The isolation rate $G$ measures coverage at a fixed distance, without any list, so two corpora with lists of the same size can still differ in $G$, as a repetitive corpus and a diverse one do.

Two extensions would make the method more useful. The first is a version for samples of accounts, since social media research often samples users and reads all of their posts. The second is a study of how the resolution that people would choose relates to the resolution chosen by large language models, which would replace our two automatic judges with a human standard.

## Limitations

The resolution $r$ is a cosine distance in the space of one encoder. The law held under both encoders on AskDocs and TIFU, but $d$ changed with the encoder, from 13.6 under Qwen3 to 18.3 under GTR on TIFU. A user must therefore fix the encoder before choosing $r$, and report the two together. Within one encoder, the same $r$ also means different things in different corpora (Section 5.6). Our calibration of $r$ rests on the judgments of two large language

models, which agree with $\kappa = 0.55$ and place the crossing at different distances, and no person checked them.

The required sample grows as $r^{-d}$, so the formula is useful only at resolutions where the required sample is a feasible number. At $d = 15$, halving $r$ multiplies the required sample by about 33,000. The formula helps plan a sample at the resolution of the question, and it cannot make a fine resolution affordable.

The constants $a$ and $b$ must be fitted for each corpus. We can explain why $b$ is below 1, but we have no expression that predicts its value.

The pilot threshold of 0.95, and the gap of a quarter between the two estimates of $d$, were both chosen on the corpora of this paper. The band of two thirds to one and a half times $n^*$ is empirical, and resampling the pilot does not reproduce it. The corpora are English social media and forum text, and the Truth Social and Bluesky collections are political collections, so their values of $d$ describe those collections.

The unit of sampling is the document. Posts by the same account are correlated, and a design that samples accounts would need a version of the estimator for clusters.

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## A Synthetic smoke test

The Zipf topic mixtures have 20,000 documents in 64 dimensions. Four settings vary the number of topics (50, 200 or 500), the Zipf exponent and the noise. Each setting was run at three seeds and eleven sample sizes from 50 to 5,000. Table 4 compares coverage $1 - G$ with the true topic mass coverage at radius $\alpha$ times the median pairwise distance of a pilot.

| $\alpha$ | Spearman $\rho$ | mean absolute error |
|---|---|---|
| 0.5 | 0.968 | 0.029 |
| 0.4 | 0.797 | 0.091 |
| 0.3 | 0.677 | 0.209 |
| 0.2 | 0.576 | 0.521 |
| 0.1 | 0.307 | 0.869 |

Table 4: Agreement of $1 - G$ with true topic coverage on the Zipf mixtures, 132 samples.

## B The sample size formula by corpus

For each pair of corpus and encoder, we fitted Equation 4 on samples of up to 1,000 documents, with five independent samples at each size. Equation 5 then gave the sample size $n^*$ that brings the isolation rate down to 30%, 20% or 10%, at every resolution where the average over the five samples reaches the target only beyond 1,000 documents. We found the true sample size by interpolating between the sample sizes we drew, on a log scale. Table 5 reports, for each pair, how many cases there were, the typical ratio of the formula's answer to the true sample size, and in how many cases the true sample size lay between two thirds and one and a half times the formula's answer (column $\leq 1.5$) or between half and twice the answer (column $\leq 2$). Two facts help in judging these numbers. Neighbouring sample sizes in our grid differ by about one and a half times, and the true sample size varies by a median of 6% across the five samples. On the six failing pairs of real text, which leave out the Zipf mixture, the true sample size lay within the narrower band in 10 of 18 cases and within the wider band in 14. When $a$ and $b$ are fitted on subsamples of one pilot of 2,000 documents instead of on draws from the whole corpus, the narrower band holds in 76 of the 83 cases on passing pairs. The wider band holds in 81 of them.

A random sample can be compared with a sample chosen for coverage. Greedy farthest point selection, the core-set construction of Sener and Savarese (2018), minimises the largest distance from any document to the chosen set. To bring the missing mass at a radius below a target, a random sample needs 0.6 to 1.0 times as many documents as greedy selection on the forum and census corpora, and 0.12 times as many on Truth Social and Bluesky. Greedy selection spends its picks on outlying documents, which are few, while the missing mass weighs documents equally.

## C Prediction of the sample that covers half the corpus

The predictions in this appendix use $d$, $a$ and $b$ from the first pilot alone. Averaging $d$ over five pilots instead moves no predicted share by more than half a percentage point.

The eighteen pairs of corpus and judge test the prediction to very different degrees. On five pairs the target is reached before the smallest sample of 100 documents, and at the point estimate of $r_{\text{same}}$ the prediction also falls below 100. On four pairs (AskDocs and TIFU under both judges) the target is reached at the 50% limit, with $G$ at 1.00. Any law that keeps $G$ near 1 predicts these four, so they

| Corpus | Enc. | cases | median | $\leq 1.5$ | $\leq 2$ |
|---|---|---|---|---|---|
| AskDocs | Qwen3 | 5 | 1.04 | 5 | 5 |
| Truth Social | Qwen3 | 6 | 0.88 | 5 | 6 |
| Census random | Qwen3 | 3 | 0.81 | 2 | 2 |
| Manifold $d$=2 | none | 9 | 1.15 | 9 | 9 |
| TIFU | Qwen3 | 6 | 1.08 | 6 | 6 |
| Bluesky | Qwen3 | 5 | 0.74 | 3 | 4 |
| TIFU | GTR | 17 | 1.24 | 17 | 17 |
| Manifold $d$=8 | none | 7 | 1.12 | 7 | 7 |
| AskDocs | GTR | 16 | 0.96 | 14 | 16 |
| Manifold $d$=4 | none | 9 | 1.09 | 9 | 9 |
| Census illegal | Qwen3 | 3 | 1.24 | 2 | 3 |
| Census covid | Qwen3 | 3 | 1.22 | 3 | 3 |
| Trump | Qwen3 | 2 | 1.14 | 2 | 2 |
| Census trump | Qwen3 | 2 | 1.56 | 1 | 2 |
| Census citizens | Qwen3 | 2 | 2.06 | 0 | 1 |
| Trump | GTR | 6 | 1.86 | 2 | 3 |
| Zipf, 200 topics | none | 5 | 69 | 0 | 0 |

Table 5: Predicted over empirical sample size for a target isolation rate, in the order of Table 1. GTR rows pool three samples of the forum corpus.

tell little about the law or about $r$. On five pairs the target is reached between 100 and 1,000 documents, inside the range of sample sizes the pilot fit used. There the predicted sample is at most a third larger or a quarter smaller than the measured one.

The remaining four pairs require the law to predict beyond the pilot, and they are the real test. On the Census covid slice under Qwen3.8, the prediction is 8% above the measured 3,356 documents. This corpus fails the pilot test, but the test checks the fit across many resolutions at once, and at a single resolution the fitted curve can still follow $G$. On Truth Social under both judges and on the random Census slice under Qwen3.8, the measured share lies near the 50% limit, between 41% and 44%. The prediction is 3.6 to 3.7 percentage points higher. Measured from the limit, the corpus stands 6 to 9 points below 50%, and the law places it only about half as far below.

The prediction is too high on all twelve rows of these two corpora, including the ends of the interval of $r_{\text{same}}$. Where both a prediction and a measurement exist, the largest errors fall at those ends. On the random Census slice the prediction is 4.5 percentage points too high, and on the Census covid slice under Gemma the law predicts 197 documents where 106 suffice. At two ends where the measurement falls below 100 documents (the Census citizens and illegal slices under Qwen3.8), the prediction is about 150 documents. Near the 50% limit the error always points toward a larger sample than needed. On the random Census slice, the full procedure of Appendix D, with $r_{\text{same}}$ also found inside the pilot, predicts 721 documents where 532 suffice.

## D Calibration inside the pilot

For five corpora, Gemma-4-31B-it judged up to 40 pairs in each distance bin, drawn inside the first pilot of 2,000 documents with the sampler, bins and prompt of Section 5.6. Table 6 gives the crossing with its pair bootstrap interval, the crossing from pairs over the whole corpus under the same judge, and the share that covers half of the corpus, predicted from the pilot's crossing with $d$, $a$ and $b$ from the same pilot, and measured at the crossing from the whole corpus. Pilots of the forum corpora hold few close pairs: AskDocs has 14 pairs closer than 0.2, and 4.5% of its resamples find no crossing, while TIFU has none closer than 0.15. The census and Truth Social pilots fill every bin above 0.1.

| Corpus | $r_{\text{same}}$, pilot | $r_{\text{same}}$, corpus | predicted | measured |
|---|---|---|---|---|
| Census random | .28 [.26, .30] | .28 | 9.5% | 7.0% |
| Truth Social | .20 [.18, .21] | .18 | 43% | 41% |
| Trump (forum) | .29 [.27, .33] | .31 | <100 docs | <100 docs |
| AskDocs | .19 [.13, .20] | .17 | 50% | 49% |
| TIFU | <.15 | .08 | 50% | 50% |

Table 6: Calibration of $r_{\text{same}}$ inside one pilot, under Gemma. TIFU: bound from the closest pilot pairs, and the prediction at the bound.

## E Probes of the collapse failure

## F The finite population correction

## G Coverage of a theme inventory

Each of the eight forum and census corpora received inventories of themes by $k$-means on the full Qwen3 embeddings at $K = 10$, 30 and 100, and two inventories with skewed masses at $K = 30$: one from $k$-means++ seeding without refinement, one by merging the 100 clusters into 30 groups of Zipf sizes. The effective number of themes (the exponential of the entropy of the masses) ranges from 6.8 to 91. For each inventory and the 1,200 samples of Section H, Table 9 gives the range across corpora of the share of theme mass present. The expected coverage $\sum_k m_k(1-(1-m_k)^n)$ matches the observed value to a mean absolute error of 0.004 over all cells. For the $K = 30$ inventory, Gemma 4 (Gemma Team, 2026) was shown each sample and the list of named themes and asked which were present, with a supporting document for each. The mass of themes it reported rose with $n$ along the

| Corpus | near duplicates | local $d$ | CV of local $d$ |
|---|---|---|---|
| AskDocs (GTR) | 0.0% | 8.1 to 16.3 | 0.15 |
| TIFU (GTR) | 0.0% | 14.4 to 21.7 | 0.10 |
| Truth Social | 0.04% | 7.9 to 15.4 | 0.16 |
| Bluesky | 0.06% | 11.5 to 18.9 | 0.13 |
| Trump (GTR) | 3.4% | 2.0 to 16.2 | 0.52 |
| Zipf, 200 topics | 0.0% | 7.8 to 18.1 | 0.21 |

Table 7: Probes of the collapse failure. Near duplicates: share of a 5,000-document sample with a neighbour within cosine distance 0.02. Local $d$: range of the TwoNN dimension over twenty $k$-means clusters, and its coefficient of variation.

| $n/N$ | $M$ (true) | $G$ | $G\,(1-n/N)$ | error |
|---|---|---|---|---|
| 0.01 | 0.762 | 0.768 | 0.760 | 0.032 |
| 0.05 | 0.590 | 0.623 | 0.592 | 0.014 |
| 0.10 | 0.497 | 0.547 | 0.492 | 0.010 |
| 0.25 | 0.347 | 0.459 | 0.344 | 0.006 |
| 0.50 | 0.198 | 0.396 | 0.198 | 0.004 |
| 0.70 | 0.110 | 0.366 | 0.110 | 0.003 |

Table 8: Missing mass of the finite corpus by sampling fraction, mean over eight Qwen3 corpora and three radii (the 25th, 50th and 75th percentiles of the pilot's nearest neighbour distances). The last column is the mean absolute error of the corrected estimate.

label coverage curve in every corpus, and the share of its reports that the labels confirm rose from 0.73 at $n = 25$ to 0.98 at $n = 400$.

| Inventory | $n$=25 | $n$=50 | $n$=100 |
|---|---|---|---|
| $k$-means, $K$=10 | 0.89 to 0.94 | 0.98 to 1.00 | 1.00 |
| $k$-means, $K$=30 | 0.58 to 0.64 | 0.82 to 0.85 | 0.94 to 0.98 |
| $k$-means, $K$=100 | 0.25 to 0.30 | 0.44 to 0.48 | 0.67 to 0.72 |
| seeding only, $K$=30 | 0.60 to 0.72 | 0.82 to 0.88 | 0.94 to 0.96 |
| Zipf merge, $K$=30 | 0.79 to 0.84 | 0.87 to 0.91 | 0.93 to 0.96 |

Table 9: Share of theme mass present in a sample, range across eight corpora.

## H Summary stability

For each of the eight forum and census corpora and each $n \in \{25, 50, 100, 200, 400\}$, thirty independent samples were summarized by Gemma 4 at temperature 0 in one paragraph of at most 150 words, and the summaries were embedded with Qwen3-Embedding-8B. Instability is the mean pairwise cosine distance among the thirty summaries. Within every corpus instability falls with $n$ and tracks $G$ (Spearman $+0.40$ to $+1.00$, seven of eight at $+0.70$ or above). Across corpora at a fixed $n$, Spearman of instability with $G$ is $+0.14$, $+0.12$, $+0.12$, $+0.07$ and $-0.17$ at $n = 25$ to 400, with bootstrap bands that include zero. TIFU illustrates why: its $G$ is near 1 at the shared radius and its

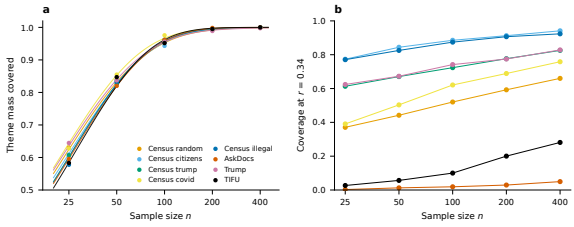



Figure 2: (a) Share of theme mass present in a sample against $n$ for the $K = 30$ inventory, eight corpora (points), with the expected coverage from each corpus's masses (lines). (b) Coverage $1 - G$ at the fixed radius $r = 0.34$ for the same samples.

summaries are the most stable, because the model summarizes it as stories of mishap whichever thirty stories it is given.

## I A bound for functions that are stable at scale $r$

For a document function $f$ with range 1 that takes the same value on documents within $r$ of each other, the estimate of the corpus mean of $f$ formed by giving each corpus document the value of its nearest sampled document errs by at most $v(r) + M(r)$, where $v(r)$ is the share of covered documents whose nearest sampled document disagrees with them on $f$. On fifty cluster indicator functions over four corpora, six sample sizes and six radii, the bound held in 144 of 144 cells and was loose for proportions (median 22 times the realised error), where the plain sample mean already errs by 0.003 to 0.07.

## J Prompts

The large language models ran on NVIDIA H100 GPUs, served with vLLM (Kwon et al., 2023) at temperature 0. Nearest neighbour search and the fits used NumPy and scikit-learn (Pedregosa et al., 2011) on CPU.

**Summary.** "Below are $\{n\}$ documents sampled from a larger collection. Write a summary of the collection in one paragraph of at most 150 words. Cover the main themes in proportion to how often they appear, name notable minority themes briefly, and do not mention individual document numbers."

**Theme naming.** "The $\{m\}$ documents below were grouped together because they are about the same theme. Give the theme a short name (at most 8 words) and a one-sentence definition that would

let a reader decide whether a new document belongs to it. Answer as JSON with keys name and definition."

**Enumeration.** "Below are $\{n\}$ documents sampled from a larger collection, followed by a list of $\{K\}$ themes found in the full collection. Decide which themes are present among these $\{n\}$ documents. A theme is present only if you can point to one document that is primarily about it; in a small sample most themes will be absent, and similar-sounding themes are distinct, so pick the single best theme for each document. Answer with one line per present theme in the form theme_id: document_number."

**Same content.** "Two documents from the same collection are shown. Judge their relation. SAME: they are about the same specific content (the same claim, question, event or story), so a reader who has read one would learn little new from the other. TOPIC: they are about the same general topic but different specific content. UNRELATED: they are about different topics."